

ASSIGNMENT BRIEF

HTU Course No: 40201341	HTU Course Name: Operating Systems
BTEC Unit Code: H/618/7486	BTEC UNIT Name: Operating Systems



Student Name/ID Number/Section	
HTU Course Number and Title	40201341 Operating Systems
BTEC Unit Code and Title	H/618/7486 Operating Systems
Academic Year	2024-2025 3
Assignment Author	Mohammad Yahia
Course Tutor	Mohammad Yahia - Asma'A Lafi
Assignment Title	Managing Modern Operating Systems
Assignment Ref No	1
Issue Date	14/08/2025
Formative Assessment dates	From 17/08/2025 to 28/08/2025
Submission Date	09/09/2025
IV Name & Date	Samir Tartir 13/08/2025

Submission Format

The Assignment consists of two parts, each with specific requirements and deadlines. All submissions must be made individually by each student.

Part 1 (In-Class Assignment):

- **Format:** Closed-book examination
- **Date:** Thursday, August 21, 2025
- **Time:** 11:00 AM – 1:00 PM

Part 2 (Take-home Assignment): This part includes the following sections:

1. Individual Written Report:

- **Format:** Microsoft Word (.docx) – **PDF submissions will not be accepted.**
- **Content:** Solutions to all assigned tasks, and clearly labeled task and subtask numbers before each answer.

2. Oral Discussion: A scheduled discussion with your instructor (and any witnesses) regarding your submitted work.

3. Signed Declaration Form: Must be included with your submission.

Submission Deadline: Tuesday, September 9, 2025, at 6:00 PM

Submission Instructions:

- Upload your report (a soft copy) to the university's eLearning platform: <https://elearning.htu.edu.jo/>
- Ensure your submission is well-formatted and complete.

Important Notes:

- No email or MS Teams submissions will be accepted.
- **Late submissions will not be considered under any circumstances.**
- Plagiarism will result in strict disciplinary action as per HTU policies.

Oral Discussion Details:

- Sessions will be scheduled by your instructor after the submission deadline.
- Attendance is mandatory at the assigned date and time.
- Sessions will be conducted in person at the university (location to be announced).
- All discussions will be recorded via MS Teams for documentation.

Unit Learning Outcomes

LO1 Investigate different operating systems, their functions and user interfaces.

LO2 Explore the processes managed by an operating system.

LO3 Demonstrate the use of different operating systems with a range of commands.

LO4 Examine how operating systems will function in the future and the implications on security.

Assignment Brief and Guidance

Task 1: An operating system (OS) serves as the backbone of a computer, managing hardware and software resources while providing essential services to users and applications. Different definitions emphasize its role as a resource manager, a control program, or a platform for application execution. Keeping in mind the various definitions and perspectives on operating systems, address the following questions in detail:

- A. Should operating systems include bundled applications such as web browsers and email clients?
 - Supporting Argument: Present a well-reasoned case in favor of including such applications with the OS. Consider aspects like user convenience, integration, and security.
 - Opposing Argument: Argue against bundling these applications, discussing potential drawbacks such as bloat, reduced flexibility, or antitrust concerns.
- B. List four critical services provided by an operating system (e.g., file management, process scheduling, memory management).
 - Explain how each service enhances user convenience and system functionality.

Task 2: Windows as an operating system evolved from a DOS shell to a universal Operating System powering PCs, servers, and cloud systems. Research the key evolutionary milestones in the development of the Windows operating system by discussing the following aspects:

- A. **GUI Innovations:** In what ways did Windows 95's Start Menu revolutionize user interaction, and how does its legacy persist in modern Windows versions?
- B. **Internet & Cloud Integration:** Evaluate the role of Internet Explorer, Edge, and cloud services in shaping Windows' functionality over time.
- C. **Security Enhancements:** Discuss the impact of features like User Account Control (UAC), Windows Defender, and TPM on OS security.
- D. **Touch & Hybrid Computing:** Why did Windows 8's touch-centric approach fail, and how did Windows 10 correct this while balancing touch and desktop usability?
- E. **'Windows as a Service' Model:** What are the advantages and disadvantages of Microsoft's shift to continuous updates instead of major version releases?"

Task 3: Consider the following page reference string:

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

How many page faults would occur for the following replacement algorithms, assuming three frames? Remember that all frames are initially empty. **You must provide detailed computations and frames at each step; otherwise, you will lose the criterion for the question:**

- A. LRU replacement
- B. FIFO replacement

Task 4: There are five processes named P1, P2, P3, P4, and P5. Their arrival time, burst time, and priority are given in the table below (1 means highest priority, while 4 means lowest priority).

Process ID	Arrival Time	Burst Time	Priority
P1	0	8	3
P2	1	1	1
P3	2	3	4
P4	3	6	3
P5	4	2	2

Draw the Gantt chart that describes process execution and calculate the average waiting time for each of the following scheduling algorithms.

- A. Round Robin Scheduling with time quantum ($q=2$)
- B. Preemptive priority scheduling.

Task 5: Osama is a new employee at TechSolutions Inc. and has been assigned to organize project files on the company's Linux server. His supervisor has given him the following tasks to complete using basic Linux commands. Help Osama by providing the appropriate **Linux commands** for each step:

- A. **List files and directories:** Osama needs to check the contents of his current working directory to see what files and folders are present. What command should he use?
- B. **Create a new directory:** He must create a directory named ProjectX.
- C. **Change directory:** He must navigate into a directory named ProjectX. What command should he run?
- D. **Print current working directory:** After moving into ProjectX, Osama wants to confirm his current location. Which command will display the full path of his present working directory?
- E. **Create a new directory:** His supervisor asks him to create a subdirectory called Backup inside ProjectX. What command should he execute?
- F. **Remove an empty directory:** Later, Osama realizes that Backup is no longer needed and decides to delete it (assuming it's empty). Which command will remove this directory?
- G. **Remove files or directories:** Osama needs to delete a non-empty folder named Old_Data along with all its contents. What command should he use to do this forcefully?
- H. **Copy files/directories:** He has to make a copy of a file named report.txt from ProjectX to the Documents directory. What command will achieve this?
- I. **Move or rename files/directories:** Osama is instructed to rename draft_v1.txt to final_draft.txt within the same directory. Which command can he use? Alternatively, how could he move this file to another directory?
- J. **Create an empty file:** Finally, Osama needs to create a new empty file named notes.txt. What command will help him do this?

Task 6: Repeat the steps from Task 5 (Linux file and directory commands) on either **Windows** (Command Prompt/PowerShell) or **macOS** (Terminal).

Important note: For each command performed in **Task 5** and **Task 6**, take a screenshot showing the command and its output as proof of correct execution.

Task 7: Modern operating systems will need to evolve significantly to meet future technological demands, particularly with advancements in Artificial Intelligence (AI). Based on this, answer the following questions:

- A. How has the integration of AI transformed the development of modern operating systems?
- B. What are the key AI technologies being incorporated into operating systems today?
- C. What role will AI-powered voice and natural language interfaces play in replacing traditional OS interaction methods?

Task 8: As the lead system administrator at a tech company, you are responsible for establishing a secure collaborative environment for the "CityTechnologies" project. Your tasks include managing user accounts, groups, file permissions, and system policies to ensure confidentiality and controlled access.

Tasks to Complete:

- A. **User Account Creation:** Create secure user accounts for employees **Khaled** and **Amal**, enforcing strong passwords and secure configurations.
- B. **Project Directory Setup:** Create a directory named **CityTechnologies** containing three files:
 - Takamul_iot
 - izam_iot
 - Shabakat_iot
- C. **Manager Account Setup:** Create an account for **ahmed_manager**, who will oversee the **CityTechnologies** project.
- D. **Engineers Group Configuration:** Create a group named **Engineers** and add **Khaled** and **Amal** as members.
- E. **Project Manager Assignment:** Assign **ahmed_manager** as the owner of the **CityTechnologies** directory and its files.
- F. **Group Ownership Assignment:** Set the group ownership of the **CityTechnologies** directory and all its files to the **Engineers** group.
- G. **Manager Access Rights:** Grant **ahmed_manager** full administrative access (read, write, execute) to all project files and directories.
- H. **Engineers Access Rights:** Provide **Khaled** and **Amal** with read, write, and execute permissions for all **CityTechnologies** files.
- I. **Restrict Other Users:** Deny access to all other users, allowing execute permissions only where necessary.
- J. **Directory Permissions:** Configure the **CityTechnologies** directory with:
 - **Owner (ahmed_manager):** Read & Write
 - **Group (Engineers):** Read-only
 - **Others:** No access (except execute if required)
- K. **Security Verification:** Confirm that all permissions align with the company's security and collaboration policies.
- L. **User Access Testing:** Log in as **Khaled** and verify proper file access and functionality.
- M. **Sensitive File Protection:** For the file **AccessControlPlan.txt** inside **CityTechnologies**:
 - **Engineers group:** Read-only
 - **ahmed_manager (owner):** Read & Write
 - **All other users:** No access
- N. **Controlled File Modifications:** Allow users to create and edit their own files but restrict them from deleting files they do not own.
- O. **Post-Project Cleanup:** After project completion, remove all user accounts (**Khaled**, **Amal**, **ahmed_manager**) and delete their home directories.

Important note: For each command performed in **Task 8**, take a screenshot showing the command and its output as proof of correct execution.

Task 9: Modern operating systems are increasingly leveraging virtualization and containerization technologies to meet the demands of future scalability in cloud computing and big data. Based on this, answer the following questions:

- A. Discuss what virtualization and containerization are.
- B. How do modern operating systems support virtualization in various ways?
- C. Why do modern systems adopt a hybrid approach combining virtualization and containerization, and how does this enhance scalability and security in environments like cloud computing?

Task 10: As cybersecurity and data privacy become increasingly critical, modern operating systems are evolving to address these challenges proactively. Based on this, answer the following questions:

- A. Evaluate the implications of open-source vs proprietary operating systems in the context of national cybersecurity strategies.
- B. Based on your evaluation of question 1, what is the best approach for national cybersecurity strategies?
- C. How do modern operating systems incorporate AI-driven threat detection to proactively mitigate future security risks?

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Investigate different operating systems, their functions and user interfaces.	P1 Summarise what an operating system is and how it works with reference to different examples. P2 Research the evolution of operating systems.	M1 Discuss the importance of operating systems.	D1 Evaluate the functionality, interface design and processes of a range of operating systems.
	P3 Research the process of Memory Management in an Operating System. P4 Investigate the process of job scheduling.		
LO2 Explore the processes managed by an operating system.	P5 Demonstrate common commands on different operating systems. P6 Compare how different commands are carried out on different operating systems.	M2 Illustrate the importance of resource management in an operating system to aid its efficiency. M3 Analyse the security of different operating systems.	D2 Evaluate the role of different operating system in meeting the needs of future technologies and the implications on security.
LO3 Demonstrate the use of different operating systems with a range of commands.	P7 Explore the core features modern operating systems will require to meet future needs.	M4 Assess how the features of modern operating systems will support the development of future needs.	
LO4 Examine how operating systems will function in the future and the implications on security.			